

Database-Traceable Uncertainty Quantification for Layered-Tissue RF Dosimetry

1 Traceable inputs



IT'IS v5.0

- 112 tissue entries
- 104 with complete dielectric data
- conductivity uncertainty: conformal interval



- thermal uncertainty: documented min/max



2 Reproducible model

plane wave

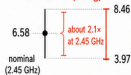


Transmission-line SAR
+
1D Pennes bioheat

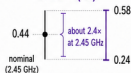
50 W/m²
0.9 / 2.45 / 5.8 GHz

3 Propagated bounds

Peak SAR (W/kg)



Peak ΔT (°C)



2.45 GHz example:

peak SAR 6.58 [3.97, 8.46] W/kg;

peak ΔT 0.44 [0.24, 0.58] °C

Cross-frequency widening across
0.9–5.8 GHz is about 2–4x.

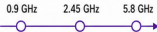
4 Dominant drivers

A. Peak SAR



Conductivity σ dominates
at all frequencies

B. Peak ΔT



Critical tissue shifts
from **muscle** to **fat**

Documented tissue-property uncertainty changes SAR and temperature margins; the controlling driver depends on endpoint and frequency.